TOPIC 3 • THEORY Advanced Mathematics • BSc Aerospace Engineering • UC3M

Second-Order Linear Equations: General Theory, Vibrations and Forced Response

Pages 5 to 8 of the class notes, in their order: the second-order linear equation and its existence theorem, superposition and the general solution, the Wronskian, constant coefficients (Cases 01, 02, 03), inhomogeneous equations with the educated guess and the example $x'' + x' - 6x = f$ solved for three forcings, and the two strategies for variable coefficients: reduction of order and variation of constants. Four slips of the notes are corrected with the check that proves them; Robinson material beyond the notes is labelled as a complement.

Ground truth: Class notes Notes.pdf, pp. 5–8 • Robinson (2004) Ch. 11 (pp. 101–110), Ch. 12 (pp. 111–119), Ch. 13 (pp. 120–130), Ch. 14 (pp. 131–140), Ch. 15 (pp. 141–149), Sec. 16.1 (p. 150), Ch. 17 (pp. 159–163), Ch. 18 (pp. 164–168) • Lecture map, Parts II–III (Sep 22 – Oct 1)

≈ 55 min study time • 56 displayed equations • 4 figures

SECTION 01 • NOTES P. 5 • ROBINSON SEC. 11.1 • LECTURE SEP 22

1. Second-order linear ODEs

SECOND-ORDER LINEAR ODES [NOTES, P. 5]

$$x''(t) + p(t)\,x'(t) + q(t)\,x(t) = f(t)$$

For example, the equation of a spring: $p(t)$ is the friction of the system, $q(t)$ the spring constant and $f(t)$ the forcing. We have a basic well-posedness (existence and uniqueness) result in the style of Picard.

With a mass m, a damping coefficient μ and a spring stiffness k, Newton's law $mx'' = -\mu x' - kx + F(t)$ divided by m is exactly this form with $p = \mu/m$, $q = k/m$ and $f = F/m$ (Section 5). Robinson starts from the general second-order equation:

Newton's second law is second order: the force fixes the acceleration $\ddot{x}$, not the velocity. A general second-order equation reads [Robinson, Eq. (11.1), p. 101]

$$\frac{d^2x}{dt^2} = f(\dot{x}, x, t). \tag{11.1}$$

1.1 Why two initial conditions

Take the simplest case $\ddot{x} = f(t)$ (11.2), the motion of a particle under a prescribed force per unit mass. Integrate twice from t_0:

1. $\dot{x}(t) - \dot{x}(t_0) = \int_{t_0}^{t} f(s)\,ds$ (Barrow's rule on $\ddot{x} = \frac{d}{dt}\dot{x}$). One constant, $\dot{x}(t_0)$, appears.
2. $x(t) - x(t_0) = \int_{t_0}^{t} \dot{x}(s)\,ds = \dot{x}(t_0)(t - t_0) + \int_{t_0}^{t}\int_{t_0}^{s} f(r)\,dr\,ds$. A second constant, $x(t_0)$, appears.

Two integrations produce two arbitrary constants, so the position alone does not determine the motion: a piece of chalk can be dropped or thrown from the same point [p. 102].

1.2 Existence and uniqueness

THEOREM [NOTES, P. 5]

Given $\ddot{x} = f(\dot{x}, x, t)$, if f, $\dfrac{\partial f}{\partial x}$ and $\dfrac{\partial f}{\partial \dot{x}}$ are continuous in a cube $\{(a_1, a_2) \times (b_1, b_2) \times (t_1, t_2)\}$, then the IVP with $x(t_0) = x_0$ and $\dot{x}(t_0) = y_0$ has a unique solution (if $t_0 \in (t_1, t_2)$, $x_0 \in (a_1, a_2)$ and $y_0 \in (b_1, b_2)$).

Robinson's statement of the same theorem, with $x_1 = x$ and $x_2 = \dot{x}$:

THEOREM 11.1 — EXISTENCE AND UNIQUENESS [ROBINSON, P. 102]

Suppose $f(x_2, x_1, t)$, $\partial f/\partial x_1$ and $\partial f/\partial x_2$ are continuous for $a_1 < x_1 < a_2, b_1 < x_2 < b_2, t_1 < t < t_2$. Then for all initial conditions

$$x(t_0) = x_0, \qquad \dot{x}(t_0) = y_0 \tag{11.4}$$

with $a_1 < x_0 < a_2, b_1 < y_0 < b_2, t_1 < t_0 < t_2$, the equation $\ddot{x} = f(\dot{x}, x, t)$ has a unique solution, with two continuous derivatives, on some interval I containing t_0.

Here x_1 stands for x and x_2 for $\dot{x}$: the theorem is Theorem 6.2 applied to the equivalent first-order system $\dot{x}_1 = x_2, \dot{x}_2 = f(x_2, x_1, t)$ (Topic 4, Theorem 25.3). For *linear* equations with continuous coefficients the solution in fact exists on the whole interval where the coefficients are continuous (a standard result, not stated in the book); with constant coefficients, for all $t \in \mathbb{R}$ (Theorem 16.1, p. 150) – no blow-up is possible.

2. Superposition, independence and the general solution

SUPERPOSITION PRINCIPLE [NOTES, P. 5]

If x_1 and x_2 are solutions to the linear homogeneous ODE $x'' + px' + qx = 0$, then any linear combination $\alpha x_1 + \beta x_2$ of x_1 and x_2 is also a solution.

Remark. If $f(t) = 0$ the equation is **homogeneous**; if $f(t) \neq 0$ it is **inhomogeneous**.

A second-order linear equation $a_2(t)\ddot{x} + a_1(t)\dot{x} + a_0(t)x = g(t)$ with $a_2(t) \neq 0$ can be divided by a_2 to give the standard form [Eqs. (11.6)-(11.7), pp. 102-103]

$$\ddot{x} + p(t)\,\dot{x} + q(t)\,x = f(t) \quad \text{(inhomogeneous)}, \qquad \ddot{x} + p(t)\,\dot{x} + q(t)\,x = 0 \quad \text{(homogeneous)}. \tag{11.6-11.7}$$

Writing $L[x] = \ddot{x} + p\dot{x} + qx$ (11.10), the equations are $L[x] = f$ and $L[x] = 0$, and L is linear (11.11):

$$L[\alpha x_1 + \beta x_2] = \alpha L[x_1] + \beta L[x_2].$$

PRINCIPLE OF SUPERPOSITION [ROBINSON, EQ. (11.8), P. 103]

If x_1 and x_2 solve the homogeneous equation (11.7), so does $x = \alpha x_1 + \beta x_2$ for any constants α, β.

PROOF

Differentiation is linear: $\dot{x} = \alpha\dot{x}_1 + \beta\dot{x}_2$ and $\ddot{x} = \alpha\ddot{x}_1 + \beta\ddot{x}_2$. Therefore

$$\ddot{x} + p\dot{x} + qx = \alpha\big(\ddot{x}_1 + p\dot{x}_1 + qx_1\big) + \beta\big(\ddot{x}_2 + p\dot{x}_2 + qx_2\big) = \alpha\cdot 0 + \beta\cdot 0 = 0.$$

■

DEFINITION — LINEAR INDEPENDENCE [NOTES, P. 5]

The functions $x_1(t), x_2(t), \dots, x_n(t)$ are **linearly independent** on an interval I if the only solution of $\alpha_1 x_1 + \alpha_2 x_2 + \cdots + \alpha_n x_n = 0$ is $\alpha_1 = \alpha_2 = \cdots = \alpha_n = 0$.

For two functions this means "not proportional": if $x_2 = cx_1$ then $cx_1 - x_2 = 0$ is a non-trivial combination; conversely, if $\alpha_1 x_1 + \alpha_2 x_2 = 0$ with $\alpha_1 \neq 0$, then $x_1 = -(\alpha_2/\alpha_1)x_2$ [pp. 104-105]. So e^{t} and e^{2t} are independent, while $\sin 2t$ and $\sin t \cos t$ are not ($\sin 2t = 2\sin t\cos t$).

THEOREM [NOTES, P. 5]

Any solution to $x'' + px' + qx = 0$ can be written as a linear combination $x(t) = C_1 x_1(t) + C_2 x_2(t)$, where x_1 and x_2 are linearly independent solutions of the same problem. By the existence and uniqueness theorem it is easy to construct two linearly independent solutions.

The example of the notes. To prove $x(t) = C_1 x_1(t) + C_2 x_2(t)$, take the IVP $x'' + px' + qx = 0$, $x(t_0) = x_0$, $x'(t_0) = y_0$, where x_0 and y_0 can vary. Imposing both conditions on $C_1 x_1 + C_2 x_2$ gives the linear system

$$\begin{pmatrix} x_1(t_0) & x_2(t_0) \\ x_1'(t_0) & x_2'(t_0) \end{pmatrix} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix} = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix},$$

which has a solution for every x_0, y_0 only when the determinant is non-zero [Notes, p. 5]. That determinant is the Wronskian of Section 3. Robinson's proof of the theorem:

PROOF

Two independent solutions exist. By Theorem 11.1 there are solutions with $x_1(t_0) = 1, \dot{x}_1(t_0) = 0$ and $x_2(t_0) = 0, \dot{x}_2(t_0) = 1$. If $x_2 = cx_1$ then $x_2(t_0) = c$ forces $c = 0$, but then $\dot{x}_2(t_0) = 0 \neq 1$. So they are not proportional.

Two are enough. Let x_1, x_2 be any two independent solutions and x any solution, with $x(t_0) = x_0$, $\dot{x}(t_0) = v_0$. We want α, β with $\alpha x_1 + \beta x_2$ matching these data:

$$\begin{pmatrix} x_1(t_0) & x_2(t_0) \\ \dot{x}_1(t_0) & \dot{x}_2(t_0) \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} x_0 \\ v_0 \end{pmatrix}. \tag{11.15}$$

This is solvable for every right-hand side if and only if the determinant $x_1(t_0)\dot{x}_2(t_0) - x_2(t_0)\dot{x}_1(t_0)$ is non-zero. Suppose instead it were zero (11.16). Then the columns are proportional: there is a constant c with $x_1(t_0) = cx_2(t_0)$ and $\dot{x}_1(t_0) = c\dot{x}_2(t_0)$ (11.17). (Robinson divides by $x_2(t_0)$ and $\dot{x}_2(t_0)$ to find c; they cannot both vanish, because then uniqueness would force $x_2 \equiv 0$, and if one vanishes the same conclusion follows directly – footnote, p. 106.) By superposition $y = cx_2$ is a solution, and it has the same data at t_0 as x_1. By uniqueness $x_1 = cx_2$ for all t, contradicting independence. Hence the determinant is non-zero, (11.15) has a solution, and by uniqueness $x = \alpha x_1 + \beta x_2$ everywhere.

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LINEAR-ALGEBRA READING (ROBINSON SEC. 11.5, STARRED)

L maps the vector space $C^2(I)$ of twice continuously differentiable functions to $C^0(I)$. The solution set of $L[x] = 0$ is its kernel, a vector space, and the theorem says $\dim \ker L = 2$ with basis $\{x_1, x_2\}$. For an n th-order equation, $\dim \ker L = n$ (Sec. 16.2).

3. The Wronskian

WRONSKIAN [NOTES, P. 5]

$$W(x_1, x_2)(t) = x_1(t)\,x_2'(t) - x_2(t)\,x_1'(t).$$

We find a solution for C_1 and C_2 if $W(x_1, x_2) \neq 0$.

3.1 What equation does W satisfy?

Let x_1 and x_2 solve $x'' + p(t)x' + q(t)x = 0$ [Notes, p. 5].

1. Product rule on both products:

$$W' = \big(x_1'x_2' + x_1x_2''\big) - \big(x_2'x_1' + x_2x_1''\big) = x_1x_2'' - x_2x_1''.$$

   The terms $x_1'x_2'$ cancel.

2. Each x_j is a solution, so $x_j'' = -px_j' - qx_j$. Substituting,

$$W' = x_1\big(-px_2' - qx_2\big) - x_2\big(-px_1' - qx_1\big) = -p\,x_1x_2' + p\,x_2x_1' - q\,x_1x_2 + q\,x_2x_1.$$

3. The q terms cancel, and $-p\,x_1x_2' + p\,x_2x_1' = -p\,(x_1x_2' - x_2x_1')$:

$$W'(t) = -p(t)\,W(t).$$

ERRATUM — CLASS NOTES, P. 5

The notes box the result as $W'(t) = p(t)\,W(t)$, without the minus sign, but their own last line before it reads $-p\,x_1x_2' + p\,x_2x_1'$, which is $-p\,W$. The solution the notes give next, $W(t) = Ce^{-\int p(t)\,dt}$, is the solution of $W' = -pW$, so only the boxed line is affected.

Check. For $x'' + x' - 6x = 0$ ($p = 1$) take $x_1 = e^{-3t}$, $x_2 = e^{2t}$: $W = e^{-3t}\cdot 2e^{2t} - e^{2t}\cdot(-3e^{-3t}) = 5e^{-t}$. Then $W' = -5e^{-t} = -1\cdot W$, while $p\,W = +5e^{-t} \neq W'$.

$W' = -pW$ is a first-order linear (and separable) equation, solved as in Topic 2: $W'/W = -p$, $\ln|W| = -\int p\,dt + c$, so

ABEL'S IDENTITY [NOTES, P. 5; ROBINSON EXERCISE 11.2]

$$W(t) = C\,e^{-\int p(t)\,dt}, \qquad \text{or, with limits,} \qquad W(t) = W(t_0)\,\exp\!\left(-\int_{t_0}^{t} p(s)\,ds\right).$$

The exponential is never zero. So, as the notes conclude (also by Picard's theorem, since $W \equiv 0$ is the solution of $W' = -pW$ through $W(t_0) = 0$): **either $W \equiv 0$ (everywhere) or $W \neq 0$ (never zero)** – either 0 for all t, or $\neq 0$ for all t. One evaluation at a single time decides independence on the whole interval. With constant p, $W(t) = W(t_0)\,e^{-p(t-t_0)}$.

3.2 Complement (Robinson): the Wronskian test

THEOREM – WRONSKIAN TEST (LECTURE, SEP 22; ROBINSON P. 107)

1. For any two differentiable functions: if $W[x_1, x_2](t_0) \neq 0$ at some $t_0 \in I$, then x_1, x_2 are linearly independent on I.
2. For two **solutions** of $\ddot{x} + p\dot{x} + qx = 0$: they are linearly independent if and only if $W \neq 0$; and then $W(t) \neq 0$ for every $t \in I$.

PROOF

(1) Suppose $\alpha x_1(t) + \beta x_2(t) = 0$ for all $t \in I$. Differentiating, $\alpha \dot{x}_1(t) + \beta \dot{x}_2(t) = 0$ too. At $t = t_0$ these two equations read $\begin{pmatrix} x_1 & x_2 \\ \dot{x}_1 & \dot{x}_2 \end{pmatrix}\begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \mathbf{0}$ with a matrix of determinant $W(t_0) \neq 0$, which is invertible; multiplying by its inverse gives $\alpha = \beta = 0$.

(2) "If": part (1). "Only if": the proof of the previous theorem showed that if $W(t_0) = 0$ at any single point (equation (11.16)) the solutions are proportional. So for independent solutions W never vanishes; for dependent ones it vanishes identically.

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THE CONVERSE OF (1) FAILS FOR ARBITRARY FUNCTIONS (ROBINSON EXERCISE 11.3)

$f_1 = t^2|t|$ and $f_2 = t^3$ are linearly independent on $\mathbb{R}$ (they agree for $t > 0$ and are opposite for $t < 0$, so no single constant relates them), yet $\dot{f}_1 = 3t|t|$ and $W = t^2|t| \cdot 3t^2 - t^3 \cdot 3t|t| = 0$ for every t. A zero Wronskian proves dependence only for solutions of a linear ODE.

4. Homogeneous equations with constant coefficients

HOMOGENEOUS LINEAR SECOND-ORDER ODES WITH CONSTANT p AND q [NOTES, P. 5]

Consider $x'' + px' + qx = 0$ for $p, q \in \mathbb{R}$. We use the previous theorem to find the general solution, since any solution can be written as a linear combination of two independent ones. We look for solutions of the type $x(t) = e^{kt}$, with k to be determined.

By the chain rule $(e^{kt})' = ke^{kt}$ and $(e^{kt})'' = k^2 e^{kt}$. Substituting:

$$(e^{kt})'' + p\,(e^{kt})' + q\,e^{kt} = k^2 e^{kt} + pk\,e^{kt} + q\,e^{kt} = \big(k^2 + pk + q\big)e^{kt} = 0.$$

Since $e^{kt} \neq 0$, the **characteristic polynomial** must vanish:

$$k^2 + pk + q = 0, \qquad k = \frac{-p \pm \sqrt{p^2 - 4q}}{2}.$$

The sign of $p^2 - 4q$ gives the three cases of the notes [Notes, p. 5].

4.1 Case 01: two distinct real roots, $p^2 > 4q$

$x_1(t) = e^{k_1 t}$ and $x_2(t) = e^{k_2 t}$ are solutions. Their Wronskian is $W = e^{k_1 t}\,k_2 e^{k_2 t} - e^{k_2 t}\,k_1 e^{k_1 t} = (k_2 - k_1)\,e^{(k_1+k_2)t} \neq 0$, so they are independent and

$$x(t) = C_1 e^{k_1 t} + C_2 e^{k_2 t}.$$

EXAMPLE 12.1 (P. 112): $\ddot{x} + \dot{x} - 6x = 0,\ x(0) = 1,\ \dot{x}(0) = 2$

$k^2 + k - 6 = (k-2)(k+3) = 0$, so $k = 2, -3$ and $x = Ae^{2t} + Be^{-3t}$, $\dot{x} = 2Ae^{2t} - 3Be^{-3t}$. At $t = 0$: $A + B = 1$, $2A - 3B = 2$. Subtracting twice the first, $-5B = 0$, so $B = 0$, $A = 1$: $x(t) = e^{2t}$.

If one root is 0 (e.g. $\ddot{x} - 2\dot{x} = 0$, $k = 0, 2$), the corresponding solution is the constant $e^{0t} = 1$: $x = A + Be^{2t}$ (Example 12.2).

4.2 Case 02: one repeated real root, $p^2 = 4q$

The root is $k = -p/2$, so $2k + p = 0$, and the notes give [Notes, p. 5]

$$x(t) = C_1 e^{kt} + C_2\, t\, e^{kt}.$$

Why te^{kt} is a solution. By the product rule, $(te^{kt})' = (1 + kt)e^{kt}$ and $(te^{kt})'' = (2k + k^2 t)e^{kt}$. Then

$$(te^{kt})'' + p\,(te^{kt})' + q\,te^{kt} = \Big[\big(k^2 + pk + q\big)\,t + \big(2k + p\big)\Big]e^{kt} = \big[0 \cdot t + 0\big]e^{kt} = 0,$$

because k is a root and $2k + p = 0$ for the double root. Wronskian: $e^{kt}(1 + kt)e^{kt} - te^{kt}\,ke^{kt} = e^{2kt} \neq 0$. Section 8.1 *derives* te^{kt} by reduction of order.

EXAMPLE 12.3 (PP. 114-115): $\ddot{x} + 2\dot{x} + x = 0,\ x(0) = 0,\ \dot{x}(0) = 1$

$(k+1)^2 = 0, k = -1$ twice. $x = (A + Bt)e^{-t}$, $\dot{x} = (B - A - Bt)e^{-t}$. Then $A = 0$ and $B - A = 1$: $x(t) = te^{-t}$.

4.3 Case 03: complex conjugate roots, $p^2 < 4q$

The roots are [Notes, p. 5]

$$k_{1,2} = -\frac{p}{2} \pm \frac{\sqrt{4q - p^2}}{2}\,i = \rho \pm \omega i, \qquad \rho = -\frac{p}{2}, \quad \omega = \frac{\sqrt{4q - p^2}}{2}.$$

1. The two exponentials are solutions (the computation above holds for complex k):

$$x(t) = C_1 e^{(\rho + \omega i)t} + C_2 e^{(\rho - \omega i)t}.$$

2. Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$, with $\cos(-\omega t) = \cos\omega t$ and $\sin(-\omega t) = -\sin\omega t$:

$$x(t) = C_1 e^{\rho t}\big(\cos\omega t + i\sin\omega t\big) + C_2 e^{\rho t}\big(\cos(-\omega t) + i\sin(-\omega t)\big) = (C_1 + C_2)\,e^{\rho t}\cos\omega t + i\,(C_1 - C_2)\,e^{\rho t}\sin\omega t.$$

3. If we choose C_1 and C_2 complex conjugates, $C_1 = \frac{1}{2}(A - iB)$ and $C_2 = \overline{C_1} = \frac{1}{2}(A + iB)$ with A, B real, then $C_1 + C_2 = A$ and $i(C_1 - C_2) = i\,(-iB) = B$ are real, and

$$x(t) = e^{\rho t}\big(A\cos\omega t + B\sin\omega t\big).$$

$\rho < 0$: decaying oscillation; $\rho = 0$: pure oscillation (simple harmonic motion); $\rho > 0$: growing oscillation. The Wronskian of $e^{\rho t}\cos\omega t$ and $e^{\rho t}\sin\omega t$ is $\omega e^{2\rho t} \neq 0$ (Robinson Exercise 11.1 (iii)).

EXAMPLE 12.4 (PP. 117-118): $\ddot{x} + 2\dot{x} + 5x = 0,\ x(0) = 1,\ \dot{x}(0) = 0$

$k = \frac{-2 \pm \sqrt{4-20}}{2} = -1 \pm 2i$, so $x = e^{-t}(A\cos 2t + B\sin 2t)$ and, by the product rule, $\dot{x} = e^{-t}\big[(2B - A)\cos 2t - (2A + B)\sin 2t\big]$. Then $A = 1, 2B - A = 0$, so $B = \frac{1}{2}$: $x(t) = e^{-t}\big(\cos 2t + \frac{1}{2}\sin 2t\big)$.

ROOTS OF $k^2 + pk + q = 0$	FUNDAMENTAL PAIR	GENERAL SOLUTION
Case 01: real, $k_1 \neq k_2$	$e^{k_1 t}$, $e^{k_2 t}$	$C_1 e^{k_1 t} + C_2 e^{k_2 t}$
Case 02: repeated, $k = -p/2$	e^{kt}, te^{kt}	$C_1 e^{kt} + C_2 te^{kt}$
Case 03: complex, $\rho \pm i\omega$	$e^{\rho t}\cos\omega t$, $e^{\rho t}\sin\omega t$	$e^{\rho t}(A\cos\omega t + B\sin\omega t)$

HIGHER ORDER: THE SAME STRATEGY (ROBINSON SEC. 16.1, LECTURE SEP 24)

For $a_n x^{(n)} + \cdots + a_1 \dot{x} + a_0 x = 0$ the trial e^{kt} gives $a_n k^n + \cdots + a_0 = 0$. Each simple real root contributes e^{kt}; a root repeated m times contributes $e^{kt}, te^{kt}, \ldots, t^{m-1}e^{kt}$; each complex pair $\rho \pm i\omega$ contributes $e^{\rho t}\cos\omega t$, $e^{\rho t}\sin\omega t$ (times $1, t, \ldots$ if repeated). The homogeneous part of Example 16.2 ($x^{(4)} - 16x = 64\sin 2t$): $x^{(4)} - 16x = 0$ has $k^4 = 16$, $k = \pm 2, \pm 2i$, so $x = Ae^{2t} + Be^{-2t} + C\cos 2t + D\sin 2t$.

SECTION 05 • ROBINSON CH. 13 • COMPLEMENT

5. Complement: the mass–spring–damper

The notes name the spring as the model behind $x'' + px' + qx = f$. Robinson develops it in full; the three cases of Section 4 become the three damping regimes.

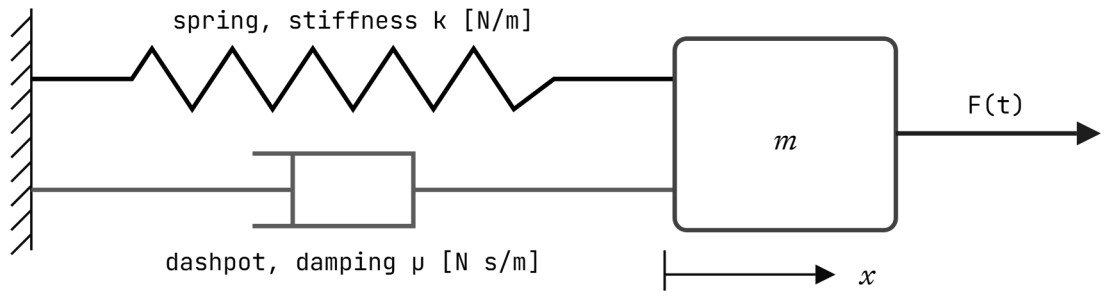


Figure 3.1. *Mass–spring–damper. The displacement x is measured from the equilibrium position, positive to the right. The spring pushes back with $-kx$ (Hooke's law), the dashpot with $-\mu\dot{x}$, and an external force $F(t)$ may act on the mass.*

5.1 Model

Hypotheses. A rigid mass m moves along one axis (one degree of freedom x, measured from equilibrium). The spring obeys Hooke's law, restoring force $-kx$; the dashpot gives a force proportional and opposite to the velocity, $-\mu\dot{x}$, with $\mu > 0$ [Robinson Sec. 13.1, p. 120; Sec. 13.3, p. 123]. Newton's second law:

$$m\ddot{x} = -kx - \mu\dot{x} + F(t) \quad \Longleftrightarrow \quad m\ddot{x} + \mu\dot{x} + kx = F(t).$$

SYMBOL	MEANING	SI UNIT
m	Mass	kg
k	Spring stiffness	N/m = kg/s²
μ	Damping coefficient	N s/m = kg/s
$\omega = \sqrt{k/m}$	Natural angular frequency	rad/s
$\lambda = \mu/m$	Damping rate	s⁻¹

Dividing by m, the free ($F = 0$) equation is [Eq. (13.6), p. 124]

$$\ddot{x} + \lambda\dot{x} + \omega^2 x = 0, \qquad \lambda = \frac{\mu}{m} > 0, \quad \omega^2 = \frac{k}{m}. \tag{13.6}$$

ERRATUM — ROBINSON P. 124
The text says "dividing by m ... and setting $\lambda = \mu/k$". Dividing $m\ddot{x} + \mu\dot{x} + kx = 0$ by m gives the coefficient μ/m for $\dot{x}$, so the correct definition is $\lambda = \mu/m$. Dimensional check: $[\mu/m] = (\text{kg/s})/\text{kg} = \text{s}^{-1}$, as required for $\lambda\dot{x}$ to have the units of $\ddot{x}$; $[\mu/k] = (\text{kg/s})/(\text{kg/s}^2) = \text{s}$ would not.

5.2 No damping: simple harmonic motion

With $\lambda = 0$: $\ddot{x} = -\omega^2 x$ (13.1), roots $\pm i\omega$, so $x = A\cos\omega t + B\sin\omega t$ (13.2). Writing $M = \sqrt{A^2 + B^2}$ and choosing ϕ with $\cos\phi = A/M$, $\sin\phi = B/M$:

$$x = M\big(\cos\phi\cos\omega t + \sin\phi\sin\omega t\big) = M\cos(\omega t - \phi). \tag{13.3}$$

The motion has amplitude M, period $2\pi/\omega = 2\pi\sqrt{m/k}$ independent of the amplitude, and lags $M\cos\omega t$ by the time ϕ/ω. The same equation governs the small oscillations of a pendulum, $\ddot{\theta} = -(g/L)\theta$, with period $2\pi\sqrt{L/g}$ (Sec. 13.2).

5.3 The three damping regimes [Sec. 13.3, pp. 124–126]

The auxiliary equation $k^2 + \lambda k + \omega^2 = 0$ has roots $k = \frac{1}{2}\big(-\lambda \pm \sqrt{\lambda^2 - 4\omega^2}\big)$.

- **Over-damped, $\lambda > 2\omega$:** two real roots $-k_1, -k_2$, both negative because $0 < \sqrt{\lambda^2 - 4\omega^2} < \lambda$. $x = Ae^{-k_1 t} + Be^{-k_2 t}$: decay with no oscillation.
- **Critically damped, $\lambda = 2\omega$:** repeated root $-\lambda/2 = -\omega$. $x = (A + Bt)e^{-\omega t}$: no oscillation, and the fastest return to equilibrium; any less damping allows oscillation.

- **Under-damped, $\lambda < 2\omega$:** roots $-\lambda/2 \pm i\sigma$ with $\sigma = \tfrac{1}{2}\sqrt{4\omega^2 - \lambda^2}$.

$$x = e^{-\lambda t/2}\big(A\cos\sigma t + B\sin\sigma t\big) = M e^{-\lambda t/2}\cos(\sigma t - \phi). \tag{13.7}$$

The amplitude decays exponentially and successive maxima are separated by the constant time $2\pi/\sigma$: setting $\dot{x} = -Me^{-\lambda t/2}\big[\tfrac{\lambda}{2}\cos(\sigma t - \phi) + \sigma\sin(\sigma t - \phi)\big] = 0$ gives $\tan(\sigma t - \phi) = -\lambda/(2\sigma)$, whose solutions are spaced by π/σ (alternating maxima and minima).

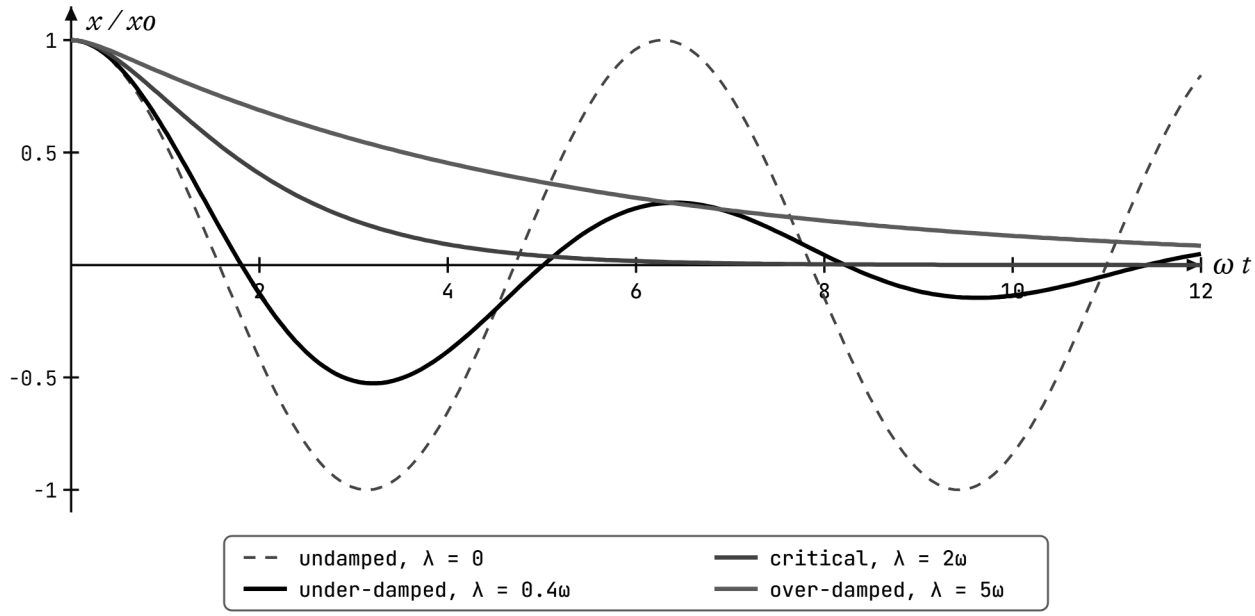


Figure 3.2. *Free response of $\ddot{x} + \lambda\dot{x} + \omega^2 x = 0$ released from rest at x_0: undamped (dashed), under-damped (indigo, decaying oscillation), critically damped (green, fastest return without overshoot) and over-damped (amber, slow creep). Compare Robinson Fig. 13.5.*

ENGINEERING NOTATION (COMPLEMENTARY)

Engineers write $\lambda = 2\zeta\omega$, i.e. $\zeta = \lambda/(2\omega) = \mu/(2\sqrt{km})$, the **damping ratio**: $\zeta < 1$ under-damped, $\zeta = 1$ critical, $\zeta > 1$ over-damped. The critical damping coefficient is $\mu_c = 2\sqrt{km}$ and the damped frequency is $\sigma = \omega\sqrt{1 - \zeta^2}$.

A LIGHTLY DAMPED STRUCTURE (ROBINSON EXERCISE 13.8, THE MILLENNIUM BRIDGE)

Model $M\ddot{x} + k\dot{x} + \lambda x = 0$ with $M \approx 4 \times 10^5$ kg, damping $k \approx 5 \times 10^4$ kg/s, stiffness $\lambda \approx 10^7$ kg/s^2 (the exercise uses k and λ the other way round from Sec. 13.3). Natural frequency $\sqrt{\lambda/M} = \sqrt{25} = 5$ rad/s, i.e. $5/(2\pi) \approx 0.8$ Hz. Critical damping would be $2\sqrt{M\lambda} = 2\sqrt{4\times10^{12}} = 4 \times 10^6$ kg/s, so the actual damping is $5 \times 10^4 / 4 \times 10^6 \approx 1.3\%$ of critical. Pedestrians walking in step added a force $\approx 300\dot{x}$ each, i.e. *negative* damping: with enough people the net coefficient of $\dot{x}$ changes sign and oscillations grow.

AEROSPACE CONNECTION: LANDING GEAR (ILLUSTRATIVE NUMBERS)

A landing-gear leg carrying $m = 2000$ kg on an oleo of stiffness $k = 2 \times 10^5$ N/m has $\omega = \sqrt{k/m} = 10$ rad/s and $\mu_c = 2\sqrt{km} = 2\sqrt{4 \times 10^8} = 4 \times 10^4$ N s/m. Designers choose ζ of the order of 0.5–0.7 to avoid repeated bouncing (under-damping) without making the strut too stiff in compression (over-damping). With $\zeta = 0.6$: $\lambda = 12\ \mathrm{s}^{-1}$, $\sigma = 10\sqrt{1 - 0.36} = 8$ rad/s, and the amplitude falls by $e^{-\lambda T/2} = e^{-6 \cdot 2\pi/8} \approx 0.009$ per damped period $T = 2\pi/\sigma$.

6. Inhomogeneous equations with constant coefficients

6.1 Structure of the general solution

SECOND-ORDER LINEAR INHOMOGENEOUS ODES WITH CONSTANT COEFFICIENTS [NOTES, P. 6]

$$x''(t) + p\,x'(t) + q\,x(t) = f(t) \tag{$*$}$$

The general solution $x(t)$ of $(*)$ is the sum of the general solution $x_H(t)$ of the associated homogeneous problem ($f \equiv 0$) and a particular solution $x_N(t)$ of $(*)$:

$$x(t) = x_H(t) + x_N(t).$$

Proof [Notes, p. 6]. Let x be any solution of $(*)$ and take $\Phi(t) = x(t) - x_N(t)$, so $\Phi' = x' - x_N'$ and $\Phi'' = x'' - x_N''$. Then

$$\Phi'' + p\Phi' + q\Phi = \big(x'' - x_N''\big) + p\big(x' - x_N'\big) + q\big(x - x_N\big) = \big(x'' + px' + qx\big) - \big(x_N'' + px_N' + qx_N\big) = f(t) - f(t) = 0.$$

Since Φ satisfies the homogeneous equation, $\Phi = x_H$ for some choice of the constants (Section 2), and $x = \Phi + x_N = x_H + x_N$.

ERRATUM — CLASS NOTES, P. 6

The last line of the proof reads "$\Phi(t) = x_H(t) = x(t) - x_N(t) \Rightarrow x(t) = x_H(t) - x_N(t)$". Solving $x_H = x - x_N$ for x gives $x = x_H + x_N$, as stated at the top of the same page. (The middle line also writes $q(x - x'_N)$ for $q(x - x_N)$.)

Check. With $f = 12$ in the example of Section 6.3, $x_N = -2$. The function $x_H - x_N$ with $x_H = 0$ is $x = 2$, and $x'' + x' - 6x = -12 \neq 12$; the function $x_H + x_N = -2$ gives $-6(-2) = 12$.

Robinson states the same result as a lemma, calling x_H the complementary function and x_N a particular integral:

LEMMA — STRUCTURE OF THE GENERAL SOLUTION [ROBINSON, SEC. 14.1, PP. 131-132]

Let $y(t) = Ay_1(t) + By_2(t)$ be the general solution of $L[y] = 0$ (the **complementary function**) and x_p any one solution of $L[x_p] = f$ (a **particular integral**). Then the general solution of $L[x] = f$ is

$$x(t) = Ay_1(t) + By_2(t) + x_p(t).$$

PROOF

It is a solution: by linearity $L[y + x_p] = L[y] + L[x_p] = 0 + f = f$. **Every solution has this form:** if $L[x] = f$, then $L[x - x_p] = L[x] - L[x_p] = f - f = 0$, so $x - x_p$ solves the homogeneous equation and equals $Ay_1 + By_2$ for some A, B (Section 2). The two constants fit any initial conditions.

■

6.2 The educated guess

To find $x_N(t)$ we make an educated guess, with trial and error, based on the form of $f(t)$; then we plug x_N into the full equation and solve for its coefficients [Notes, p. 6].

RULES OF THE NOTES [NOTES, P. 6]

- **Polynomial** of order n: $x_N(t) = C_n t^n + C_{n-1} t^{n-1} + \cdots + C_2 t^2 + C_1 t + C_0$ (all powers).
- $f(t) = Ce^{\alpha t}$**.** Case 1: if α is not a root of $k^2 + pk + q = 0$, then $x_N(t) = Ae^{\alpha t}$. Case 2: if α is a root, then $x_N(t) = Ate^{\alpha t}$.
- $f(t) = \alpha \cos \omega t + \beta \sin \omega t$**:** $x_N(t) = A \cos \omega t + B \sin \omega t$; if it is already in $x_H(t)$, then $x_N(t) = At \cos \omega t + Bt \sin \omega t$.
- **Sums:** if f is the sum of two or more terms, solve for each term and add ("decomposition").
- **Higher order:** the procedure also works for higher-order equations: get the n roots, build x_H from terms $C_n e^{k_n t}$, and find x_N with the same guesses.

Why the rules work. Derivatives of a polynomial of degree n are polynomials of degree $\leq n$, derivatives of $e^{\alpha t}$ are multiples of $e^{\alpha t}$, and derivatives of $\cos \omega t$, $\sin \omega t$ are combinations of the same two functions. So $x''_N + px'_N + qx_N$ has the same form as the guess, and matching coefficients gives linear equations. If a term

of the guess solves the homogeneous equation it contributes 0 and cannot match f: that is why the guess is multiplied by t ("Case 2", resonance).

PRECISION (ROBINSON SEC. 14.3 AND 14.5)
If α is a *double* root of $k^2 + pk + q = 0$, both $e^{\alpha t}$ and $te^{\alpha t}$ solve the homogeneous equation and the guess must be $At^2e^{\alpha t}$. Robinson's rule of thumb covers every case: multiply by t until no term of the guess solves the homogeneous equation. The table below collects his guesses.

FORCING $f(t)$	STANDARD GUESS x_p	IF A TERM OF THE GUESS SOLVES $L[y] = 0$
Polynomial of degree n	$c_n t^n + \cdots + c_1 t + c_0$ (all powers)	Multiply by t (e.g. root $k = 0$)
$ce^{\kappa t}$	$Ce^{\kappa t}$	$Cte^{\kappa t}$; $Ct^2e^{\kappa t}$ if κ is a repeated root
$\alpha \sin \sigma t + \beta \cos \sigma t$	$C \sin \sigma t + D \cos \sigma t$ (both, always)	$t(C \sin \sigma t + D \cos \sigma t)$ – resonance
Sum $\alpha f_1 + \beta f_2$	$\alpha x_1 + \beta x_2$ with $L[x_j] = f_j$	Treat each part separately (14.7)
Products, e.g. $te^{-t} \cos 2t$	Product of the guesses: $(At + B)e^{-t} \cos 2t + (Ct + D)e^{-t} \sin 2t$	As above

RULE OF THUMB [ROBINSON, SEC. 14.5, P. 139]
Take the standard guess. If it contains terms that satisfy the homogeneous equation, multiply the guess by t; repeat until no term of the guess solves the homogeneous equation.

6.3 The example of the notes: $x'' + x' - 6x = f(t)$

Step 1: find $x_H(t)$ [Notes, p. 6]. Here $p = 1, q = -6$. The characteristic polynomial is $k^2 + pk + q = k^2 + k - 6 = (k + 3)(k - 2) = 0$, so $k = -3, 2$ (Case 01) and

$$x_H(t) = C_1 e^{-3t} + C_2 e^{2t}.$$

Step 2: find $x_N(t)$ for three forcings.

(A) $f(t) = 12$ [NOTES, P. 6]

1. A constant is a polynomial of degree 0: define $x_N(t) = C$. Then $x_N' = 0$ and $x_N'' = 0$.
2. Plug in: $0 + 0 - 6C = 12$, so $C = -2$ and $x_N(t) = -2$.
3. General solution: $x(t) = C_1 e^{-3t} + C_2 e^{2t} - 2$.

Check: $(-2)'' + (-2)' - 6(-2) = 12$. As the constant is the equilibrium of the forced equation, $x \to -2$ only along $C_2 = 0$; any $C_2 \neq 0$ makes e^{2t} dominate.

(B) $f(t) = 36t$ [NOTES, P. 6]

1. Degree-1 polynomial: $x_N(t) = a_1 t + a_0$ (the notes call the coefficients C_1, C_0; other letters avoid a clash with the constants of x_H). Then $x_N' = a_1$ and $x_N'' = 0$.
2. Plug in: $0 + a_1 - 6(a_1 t + a_0) = 36t$, i.e. $-6a_1\, t + (a_1 - 6a_0) = 36t + 0$.
3. Match the coefficient of t: $-6a_1 = 36$, so $a_1 = -6$. Match the constants: $a_1 - 6a_0 = 0$, so $a_0 = a_1/6 = -1$.
4. $x_N(t) = -6t - 1$ and $x(t) = C_1 e^{-3t} + C_2 e^{2t} - 6t - 1$.

Check: $0 + (-6) - 6(-6t - 1) = -6 + 36t + 6 = 36t$. (The notes write the first matching as $6C_1 t = 36t$ before concluding $C_1 = -6$; the coefficient of t is $-6C_1$, and the value -6 is right.)

(C) $f(t) = 4e^{-2t}$ [NOTES, P. 6]

1. $\alpha = -2$ is not a root ($k = -3, 2$): Case 1, $x_N = Ae^{-2t}$, with $x_N' = -2Ae^{-2t}$ and $x_N'' = 4Ae^{-2t}$ (chain rule).
2. Plug in: $4Ae^{-2t} - 2Ae^{-2t} - 6Ae^{-2t} = -4Ae^{-2t} = 4e^{-2t}$, so $A = -1$.
3. $x_N(t) = -e^{-2t}$ and $x(t) = C_1 e^{-3t} + C_2 e^{2t} - e^{-2t}$.

Check: $(-4 + 2 + 6)\, e^{-2t} = 4e^{-2t}$.

DECOMPOSITION: $f(t) = 12 + 36t + 4e^{-2t}$

By linearity the particular solutions add: $x_N = -2 + (-6t - 1) + (-e^{-2t}) = -6t - 3 - e^{-2t}$. Check: $x_N' = -6 + 2e^{-2t}$, $x_N'' = -4e^{-2t}$, and $x_N'' + x_N' - 6x_N = (-4 + 2 + 6)e^{-2t} - 6 + 36t + 18 = 4e^{-2t} + 12 + 36t$.

6.4 Complement (Robinson): more examples

EXAMPLE 14.4 (P. 134): POLYNOMIAL FORCING, $\ddot{x} + \dot{x} - 6x = 216t^3$

The complementary function is $Ae^{2t} + Be^{-3t}$. Try the full cubic $x_p = Ct^3 + Dt^2 + Et + F$, so $\dot{x}_p = 3Ct^2 + 2Dt + E$ and $\ddot{x}_p = 6Ct + 2D$. Substituting and collecting powers:

$$-6Ct^3 + (3C - 6D)t^2 + (6C + 2D - 6E)t + (2D + E - 6F) = 216t^3.$$

Matching coefficients from the top: $-6C = 216 \Rightarrow C = -36$; $3C - 6D = 0 \Rightarrow D = -18$; $6C + 2D - 6E = 0 \Rightarrow E = (-216 - 36)/6 = -42$; $2D + E - 6F = 0 \Rightarrow F = (-36 - 42)/6 = -13$. Hence $x = Ae^{2t} + Be^{-3t} - 36t^3 - 18t^2 - 42t - 13$. A bare Ct^3 would fail: its derivatives generate t^2, t and constant terms that must be cancelled.

EXAMPLE 14.8 (P. 136): EXPONENTIAL FORCING THAT HITS THE COMPLEMENTARY FUNCTION, $\ddot{x} + \dot{x} - 6x = 5e^{-3t}$

e^{-3t} already solves the homogeneous equation, so Ce^{-3t} gives $0 = 5e^{-3t}$. Multiply by t: $x_p = Cte^{-3t}$, with $\dot{x}_p = C(1 - 3t)e^{-3t}$ and $\ddot{x}_p = C(-6 + 9t)e^{-3t}$ (product rule twice). Then

$$\ddot{x}_p + \dot{x}_p - 6x_p = C\big[(-6+9t) + (1-3t) - 6t\big]e^{-3t} = -5Ce^{-3t} = 5e^{-3t} \;\Rightarrow\; C = -1.$$

The te^{-3t} terms cancel, as they must. $x = Ae^{2t} + Be^{-3t} - te^{-3t}$.

EXAMPLE 14.10 (PP. 137–138): SINUSOIDAL FORCING, $\ddot{x} + 2\dot{x} + x = 100\cos 2t$

Complementary function $(A + Bt)e^{-t}$. Try $x_p = C\sin 2t + D\cos 2t$ (both terms: the $\dot{x}$ term turns cosines into sines). $\dot{x}_p = 2C\cos 2t - 2D\sin 2t$, $\ddot{x}_p = -4C\sin 2t - 4D\cos 2t$, so

$$\ddot{x}_p + 2\dot{x}_p + x_p = (4C - 3D)\cos 2t - (3C + 4D)\sin 2t = 100\cos 2t.$$

$3C + 4D = 0$ and $4C - 3D = 100$ give $C = 16$, $D = -12$ (check: $64 + 36 = 100$, $48 - 48 = 0$). So $x = (A + Bt)e^{-t} + 16\sin 2t - 12\cos 2t$. As $t \to \infty$ the transient dies and the steady oscillation has amplitude $\sqrt{16^2 + 12^2} = 20$.

EXAMPLE 14.11 (P. 138): RESONANCE, $\ddot{x} + x = 8\cos t$

$\cos t$ and $\sin t$ solve the homogeneous equation, so try $x_p = t(C\sin t + D\cos t)$. Differentiating twice with the product rule, $\ddot{x}_p = 2C\cos t - 2D\sin t - t(C\sin t + D\cos t)$, so $\ddot{x}_p + x_p = 2C\cos t - 2D\sin t = 8\cos t$: $C = 4$, $D = 0$. $x = A\sin t + B\cos t + 4t\sin t$: the forced part grows linearly in time.

SECTION 07 • ROBINSON CH. 15 • COMPLEMENT

7. Complement: forced vibrations and resonance

The resonance rule of the notes ("if it is already in x_H, multiply by t") has a physical meaning, developed by Robinson.

Force the oscillator at angular frequency α: $\ddot{x} + \omega^2 x = a\cos\alpha t$ (15.3).

- $\alpha \neq \omega$ [Sec. 15.1.1]: try $x_p = C\cos\alpha t$ (no sine needed – there is no $\dot{x}$ term): $(-\alpha^2 + \omega^2)C = a$, so $x = M\cos(\omega t - \phi) + \dfrac{a}{\omega^2 - \alpha^2}\cos\alpha t$ (15.4). Bounded, but large when $\alpha \approx \omega$. From rest, $x = \dfrac{\cos\alpha t - \cos\omega t}{\omega^2 - \alpha^2} \cdot a = \dfrac{2a}{\omega^2 - \alpha^2}\sin\dfrac{(\omega+\alpha)t}{2}\sin\dfrac{(\omega-\alpha)t}{2}$: a fast oscillation modulated by a slow envelope – **beats** (Exercises 15.1–15.2).
- $\alpha = \omega$ [Sec. 15.1.2]: the rule of thumb gives $x_p = \dfrac{a}{2\omega}\,t\sin\omega t$: the amplitude grows without bound. This is **resonance** (Robinson recalls the 1831 Broughton bridge collapse under marching troops). The same result follows from the beats formula by letting $\alpha \to \omega$ with L'Hôpital's rule (Ex. 15.3).

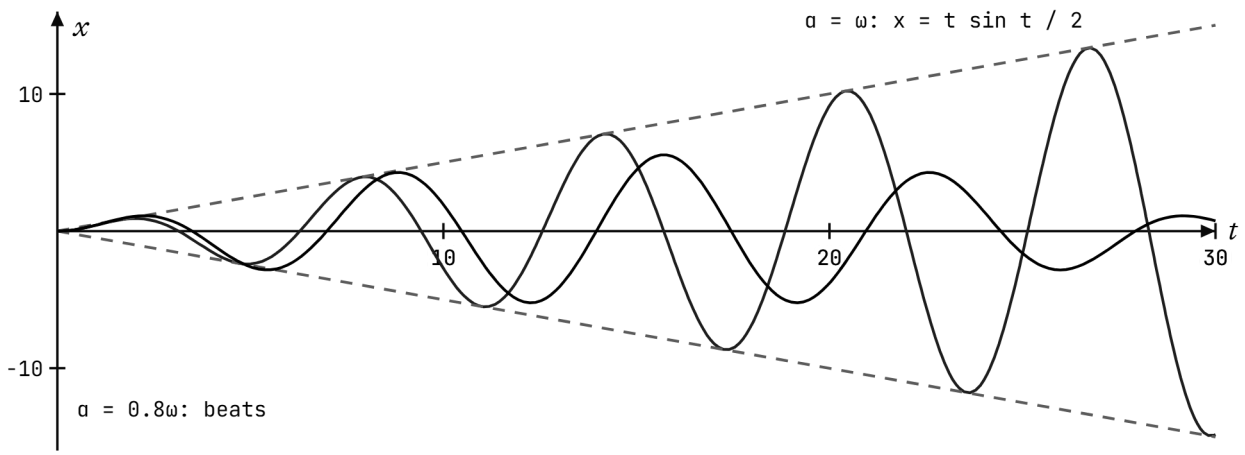


Figure 3.3. *Undamped forcing $\ddot x+\omega^2 x=\cos\alpha t$, $\omega=1$, from rest. Red: exact resonance $\alpha=\omega$, amplitude growing linearly, $x=t\sin t/2$ (Robinson Sec. 15.1.2, Ex. 15.3). Indigo: $\alpha=0.8$, bounded beats $x=(\cos\alpha t-\cos\omega t)/(\omega^2-\alpha^2)$ (Ex. 15.1–15.2).*

7.1 Damped forcing: the frequency response [Sec. 15.2, pp. 145–148]

For $\ddot x+\lambda\dot x+\omega^2 x=a\cos\alpha t$ with $\lambda>0$, try $x_p=C\sin\alpha t+D\cos\alpha t$. Matching sines and cosines:

$$C(\omega^2-\alpha^2)=D\lambda\alpha,\qquad D(\omega^2-\alpha^2)+C\lambda\alpha=a.$$

Solving (substitute C from the first into the second):

$$C=\frac{a\lambda\alpha}{(\omega^2-\alpha^2)^2+\lambda^2\alpha^2},\qquad D=\frac{a(\omega^2-\alpha^2)}{(\omega^2-\alpha^2)^2+\lambda^2\alpha^2},\qquad x_p=C\sin\alpha t+D\cos\alpha t. \tag{15.5}$$

The complementary function decays (all roots have negative real part), so after a transient only x_p remains, with amplitude

$$R(\alpha)=\sqrt{C^2+D^2}=\frac{a}{\sqrt{(\omega^2-\alpha^2)^2+\lambda^2\alpha^2}}=a\,F(\alpha),\qquad F(\alpha)=\frac{1}{\sqrt{(\omega^2-\alpha^2)^2+\lambda^2\alpha^2}}. \tag{15.6}$$

The denominator is smallest where $\frac{d}{d\alpha}\left[(\omega^2-\alpha^2)^2+\lambda^2\alpha^2\right]=-4\alpha(\omega^2-\alpha^2)+2\lambda^2\alpha=0$ (chain rule on the square), i.e. $2\alpha\,(2\alpha^2-2\omega^2+\lambda^2)=0$. For $\lambda^2<2\omega^2$ this gives an interior peak at $\alpha^2=\omega^2-\tfrac12\lambda^2$, where $R_{\max}=\dfrac{2a}{\lambda\sqrt{4\omega^2-\lambda^2}}\approx\dfrac{a}{\lambda\omega}$ for small λ. The response is finite for any $\lambda>0$ but inversely proportional to the damping: **pseudo-resonance**. For $\lambda^2\ge 2\omega^2$ there is no peak: $R(\alpha)$ decreases monotonically from $R(0)=a/\omega^2$.

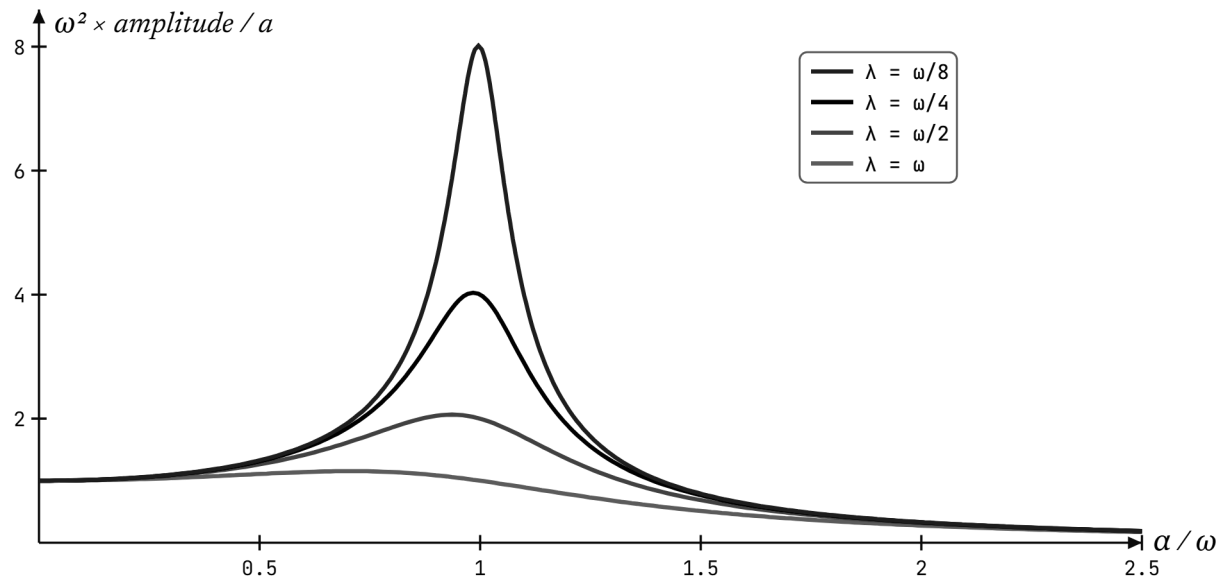


Figure 3.4. *Steady-state amplitude of $\ddot{x} + \lambda\dot{x} + \omega^2 x = a\cos\alpha t$, scaled by a/ω^2: $\omega^2 F(\alpha) = \omega^2/\sqrt{(\omega^2-\alpha^2)^2 + \lambda^2\alpha^2}$ (Robinson (15.6), Fig. 15.5). The peak sharpens and moves towards $\alpha = \omega$ as damping decreases.*

- **Limits:** $\alpha \to 0$: $R \to a/\omega^2$ (static deflection under a steady force); $\alpha \to \infty$: $R \approx a/\alpha^2 \to 0$ (inertia dominates).
- **Dimensions:** a is force per unit mass [m/s²]; $[\omega^2 - \alpha^2] = [\lambda\alpha] = \mathrm{s}^{-2}$, so $[R] = \mathrm{m}$.

AEROSPACE CONNECTION

Rotor blades, engine mounts and wing structures are tuned so that their natural frequencies stay away from forcing frequencies such as rotor or shaft speeds. Helicopter *ground resonance* and aeroelastic *flutter* are more dangerous still: as in the Millennium Bridge example, the coupling with the excitation acts as negative damping, and the amplitude grows exponentially rather than linearly.

SECTION 08 • NOTES P. 7 • ROBINSON CH. 17 • LECTURE SEP 24–29

8. Variable coefficients: reduction of order

SECOND-ORDER LINEAR ODES WITH VARIABLE COEFFICIENTS [NOTES, P. 7]

$$x''(t) + p(t)\,x'(t) + q(t)\,x(t) = f(t)$$

In general we will not be able to find all solutions, but we will study two different strategies.

8.1 Strategy 1: reduction of order

This method finds a second solution of a homogeneous problem when one is already known [Notes, p. 7]. Suppose $u(t)$ solves $x'' + p(t)x' + q(t)x = 0$, and look for a second solution of the form $x(t) = y(t)\,u(t)$.

1. Product rule: $x' = y'u + yu'$, and differentiating again, $x'' = y''u + y'u' + y'u' + yu'' = y''u + 2y'u' + yu''$.

2. Plug x into the equation:

$$\big[y''u + 2y'u' + yu''\big] + p(t)\big[y'u + yu'\big] + q(t)\big[yu\big] = 0.$$

3. Group by y, y' and y'':

$$\big[u'' + p(t)u' + q(t)u\big]\,y + \big[2u' + p(t)u\big]\,y' + u\,y'' = 0.$$

   The first bracket is 0 because u is a solution, so y itself disappears.

4. Define $z = y'$ [Notes, p. 7]: z satisfies the **first-order** equation

$$u\,z' + \big[2u' + p(t)\,u\big]\,z = 0.$$

The notes stop here. To finish (Robinson Exercise 17.7), the equation is separable: where $u \neq 0$ and $z \neq 0$, $\dfrac{z'}{z} = -\dfrac{2u'}{u} - p$, so $\ln|z| = -2\ln|u| - \int p\,dt + c$ and

$$z = \frac{A\,e^{-\int p\,dt}}{u^2}, \qquad y = \int z\,dt, \qquad x_2 = u\int \frac{e^{-\int p\,dt}}{u^2}\,dt.$$

> **CASE 02 OF SECTION 4 DERIVED**
>
> For $x'' + px' + qx = 0$ with a double root $k = -p/2$, take $u = e^{kt}$. Then $e^{-\int p\,dt} = e^{-pt} = e^{2kt}$ and $u^2 = e^{2kt}$, so $z = A$, $y = At + B$ and $x = (At + B)\,e^{kt}$: the second solution te^{kt} of Case 02 comes out of the method.

Robinson writes the same computation for $a(t)\ddot{x} + b(t)\dot{x} + c(t)x = 0$:

1. Substitute $x = u(t)\,y(t)$ (the constant A in Au is promoted to a function). Product rule: $\dot{x} = \dot{u}y + u\dot{y}$, $\ddot{x} = \ddot{u}y + 2\dot{u}\dot{y} + u\ddot{y}$.

2. Substitute and group by $y, \dot y, \ddot y$:

$$y\,\big[a\ddot u + b\dot u + cu\big] + a\big(2\dot u\dot y + u\ddot y\big) + bu\dot y = 0.$$

The bracket is zero because u is a solution – this is why the method works.

3. y itself has disappeared. With $z = \dot y$ we get the first-order linear equation $\big[au\big]\dot z + \big[2a\dot u + bu\big]z = 0$ (17.3).

4. Solve for z (separable or integrating factor), integrate once to get y, and set $x = uy$.

In the normalised form $\ddot x + p\dot x + qx = 0$ the result is $z = \dfrac{A\,e^{-\int p\,dt}}{u^2}$ (Robinson Exercise 17.7), so $x_2 = u\displaystyle\int \frac{e^{-\int p\,dt}}{u^2}\,dt$. Robinson advises applying the method rather than memorising this formula.

EXAMPLE 17.1 (PP. 160-161): DERIVING THE REPEATED-ROOT SOLUTION

For $\ddot x - 2\lambda\dot x + \lambda^2 x = 0$ with known $u = e^{\lambda t}$, set $x = ye^{\lambda t}$: $\dot x = (\lambda y + \dot y)e^{\lambda t}$, $\ddot x = (\lambda^2 y + 2\lambda\dot y + \ddot y)e^{\lambda t}$. Substituting and cancelling $e^{\lambda t} \neq 0$:

$$\lambda^2 y + 2\lambda\dot y + \ddot y - 2\lambda^2 y - 2\lambda\dot y + \lambda^2 y = \ddot y = 0 \implies y = A + Bt \implies x = (A + Bt)e^{\lambda t}. \quad \boxed{(17.4)}$$

The second solution $te^{\lambda t}$ of Section 4.2 is now derived, not guessed.

EXAMPLE 17.2 (PP. 161-162): $t^2\ddot x - 2t\dot x - 4x = 0$, $t > 0$, WITH $u = 1/t$

1. Check u: $\dot u = -t^{-2}$, $\ddot u = 2t^{-3}$, so $t^2(2t^{-3}) - 2t(-t^{-2}) - 4t^{-1} = (2 + 2 - 4)t^{-1} = 0$.
2. $x = y/t$: $\dot x = \dfrac{\dot y}{t} - \dfrac{y}{t^2}$, $\ddot x = \dfrac{\ddot y}{t} - \dfrac{2\dot y}{t^2} + \dfrac{2y}{t^3}$.
3. Substitute: $t\ddot y - 2\dot y + \dfrac{2y}{t} - 2\dot y + \dfrac{2y}{t} - \dfrac{4y}{t} = t\ddot y - 4\dot y = 0$.
4. $z = \dot y$: $\dot z - \dfrac{4}{t}z = 0$. Integrating factor $e^{-4\ln t} = t^{-4}$: $\frac{d}{dt}(t^{-4}z) = 0$, so $z = ct^4$.
5. $y = \int ct^4\,dt = \dfrac{c}{5}t^5 + b$ and $x = y/t = at^4 + b/t$. The new independent solution is $x_2 = t^4$.

Check: $t^2(12t^2) - 2t(4t^3) - 4t^4 = (12 - 8 - 4)t^4 = 0$. Wronskian: $t^{-1}\cdot 4t^3 - t^4\cdot(-t^{-2}) = 5t^2 \neq 0$ for $t > 0$.

9. Strategy 2: variation of constants

VARIATION OF CONSTANTS [NOTES, P. 8]

To find a particular solution of the complete equation $x'' + p(t)x' + q(t)x = f(t)$ when two linearly independent solutions x_1, x_2 of the homogeneous equation are known. **Idea:** look for a solution $x(t) = u_1(t)\,x_1(t) + u_2(t)\,x_2(t)$, where we need to solve for u_1 and u_2.

9.1 Derivation, step by step

1. **First derivative** (product rule on each term):

$$x' = u_1' x_1 + u_1 x_1' + u_2' x_2 + u_2 x_2'.$$

2. **Impose** $u_1' x_1 + u_2' x_2 = 0$. Two unknown functions and one equation leave one free condition; this choice keeps u_j'' out of x''. Where $x_1 \neq 0$ it gives

$$u_1' = -u_2' \frac{x_2}{x_1}. \qquad (*)$$

Then $x' = u_1 x_1' + u_2 x_2'$.

3. **Second derivative:** $x'' = u_1' x_1' + u_1 x_1'' + u_2' x_2' + u_2 x_2''$.

4. **Plug in** and regroup by u_1 and u_2:

$$\big(u_1' x_1' + u_1 x_1'' + u_2' x_2' + u_2 x_2''\big) + p\big(u_1 x_1' + u_2 x_2'\big) + q\big(u_1 x_1 + u_2 x_2\big) = f,$$

$$u_1\big(x_1'' + p x_1' + q x_1\big) + u_2\big(x_2'' + p x_2' + q x_2\big) + u_1' x_1' + u_2' x_2' = f.$$

Both brackets are 0 because x_1 and x_2 satisfy the homogeneous equation, so

$$u_1' x_1' + u_2' x_2' = f(t).$$

5 **Substitute $(*)$:**

$$\left(-u_2'\frac{x_2}{x_1}\right)x_1' + u_2'x_2' = u_2'\left(x_2' - \frac{x_2x_1'}{x_1}\right) = u_2'\,\frac{x_1x_2' - x_2x_1'}{x_1} = u_2'\,\frac{W}{x_1} = f.$$

6 **Solve:** $u_2' = \dfrac{f\,x_1}{W}$, and from $(*)$, $u_1' = -\dfrac{f\,x_1}{W}\cdot\dfrac{x_2}{x_1} = -\dfrac{f\,x_2}{W}$, with $W = x_1x_2' - x_2x_1'$.

VARIATION OF CONSTANTS

$$u_1' = -\frac{f\,x_2}{W}, \qquad u_2' = \frac{f\,x_1}{W}, \qquad W = x_1x_2' - x_2x_1',$$

$$u_1 = -\int \frac{x_2\,f}{W}\,dt, \qquad u_2 = \int \frac{x_1\,f}{W}\,dt, \qquad x_N = u_1x_1 + u_2x_2.$$

ERRATUM — CLASS NOTES, P. 8: THE SIGN OF u_1'

The notes write $u_1' = \dfrac{f\,x_2}{W}$ and $u_1 = \displaystyle\int \frac{x_2 f}{W}$, without the minus sign. Their own relation $(*)$, $u_1' = -u_2'x_2/x_1$, with $u_2' = fx_1/W$, gives $u_1' = -fx_2/W$. Robinson's formula (18.8) has the minus sign too.

Check with the example of Section 6.3, $f = 12$. $x_1 = e^{-3t}$, $x_2 = e^{2t}$, $W = 5e^{-t}$ (Section 3).

- $u_2' = \dfrac{12\,e^{-3t}}{5e^{-t}} = \dfrac{12}{5}e^{-2t}$, so $u_2 = -\dfrac{6}{5}e^{-2t}$ and $u_2x_2 = -\dfrac{6}{5}$.
- Correct sign: $u_1' = -\dfrac{12\,e^{2t}}{5e^{-t}} = -\dfrac{12}{5}e^{3t}$, so $u_1 = -\dfrac{4}{5}e^{3t}$, $u_1x_1 = -\dfrac{4}{5}$ and $x_N = -\dfrac{4}{5} - \dfrac{6}{5} = -2$ (the result of Section 6.3).
- Sign of the notes: $u_1 = +\dfrac{4}{5}e^{3t}$ gives $x_N = \dfrac{4}{5} - \dfrac{6}{5} = -\dfrac{2}{5}$, and $-6\cdot\left(-\dfrac{2}{5}\right) = \dfrac{12}{5} \neq 12$.

9.2 Complement (Robinson): the same result without dividing by x_1

Step 2 divides by x_1, which may vanish (for instance $x_1 = \sin t$ at $t = 0$). Robinson solves the two conditions as a linear system with Cramer's rule, which needs only $W \neq 0$ and gives the same formulas [Robinson, Ch. 18, pp. 164-166].

The two conditions of steps 2 and 4 form a linear system for u_1', u_2':

$$\begin{pmatrix} x_1 & x_2 \\ x_1' & x_2' \end{pmatrix}\begin{pmatrix} u_1' \\ u_2' \end{pmatrix} = \begin{pmatrix} 0 \\ f \end{pmatrix}.$$

Its determinant is the Wronskian $W = x_1x_2' - x_2x_1'$, never zero for independent solutions (Section 3). Cramer's rule:

$$u_1' = \frac{\begin{vmatrix} 0 & x_2 \\ f & x_2' \end{vmatrix}}{W} = -\frac{x_2\, f}{W}, \qquad u_2' = \frac{\begin{vmatrix} x_1 & 0 \\ x_1' & f \end{vmatrix}}{W} = \frac{x_1\, f}{W}.$$

Robinson writes the forcing as $g(t)$:

VARIATION OF CONSTANTS FORMULA [ROBINSON, EQ. (18.8), P. 166]

$$x_p(t) = -x_1(t) \int \frac{x_2(t)\, g(t)}{W(t)}\, dt + x_2(t) \int \frac{x_1(t)\, g(t)}{W(t)}\, dt. \tag{18.8}$$

Constants of integration only add multiples of x_1, x_2, which belong to the complementary function.

EXAMPLE 18.1 (PP. 166-167): $\ddot{x} + \dot{x} - 6x = 5e^{-3t}$ AGAIN

$x_1 = e^{-3t}, x_2 = e^{2t}, W = e^{-3t} \cdot 2e^{2t} - e^{2t} \cdot (-3e^{-3t}) = 5e^{-t}$. Then $\dfrac{x_2 g}{W} = \dfrac{e^{2t} \cdot 5e^{-3t}}{5e^{-t}} = 1$ and $\dfrac{x_1 g}{W} = \dfrac{e^{-3t} \cdot 5e^{-3t}}{5e^{-t}} = e^{-5t}$, so

$$x_p = -e^{-3t} \int 1\, dt + e^{2t} \int e^{-5t}\, dt = -te^{-3t} + e^{2t}\left(-\frac{e^{-5t}}{5}\right) = -te^{-3t} - \frac{1}{5}e^{-3t}.$$

The second term is a multiple of x_1 and is absorbed into Ae^{-3t}: the particular integral $-te^{-3t}$ of Example 14.8 is recovered.

ERRATUM — ROBINSON P. 166

The book writes the second integral as $5e^{2t} \int e^{-5t} dt$ and the result as $-te^{-3t} - e^{-3t}$. The factor 5 was already cancelled by $W = 5e^{-t}$, so the term is $e^{2t} \cdot (-\frac{1}{5}e^{-5t}) = -\frac{1}{5}e^{-3t}$. Check: $\frac{d}{dt}\big(-\frac{1}{5}e^{-5t}\big) = e^{-5t}$, so $\int e^{-5t} dt = -\frac{1}{5}e^{-5t}$. The slip is harmless for the general solution, since any multiple of e^{-3t} is absorbed into the arbitrary constant, as Robinson notes.

EXAMPLE 18.2 (PP. 167-168): $\ddot{x} + x = \tan t$ — BEYOND UNDETERMINED COEFFICIENTS

1. $x_1 = \sin t, x_2 = \cos t$; ansatz $x = u \sin t + v \cos t$ with $\dot{u} \sin t + \dot{v} \cos t = 0$ (18.10).
2. Then $\dot{x} = u \cos t - v \sin t$, $\ddot{x} = \dot{u} \cos t - u \sin t - \dot{v} \sin t - v \cos t$, and the equation reduces to $\dot{u} \cos t - \dot{v} \sin t = \tan t$ (18.11).
3. Multiply (18.10) by $\sin t$, (18.11) by $\cos t$ and add: $\dot{u} = \tan t \cos t = \sin t$. Multiply (18.10) by $\cos t$, (18.11) by $\sin t$ and subtract: $\dot{v} = -\dfrac{\sin^2 t}{\cos t} = \cos t - \sec t$.
4. Integrate: $u = -\cos t$; $v = \sin t - \ln|\sec t + \tan t|$, using $\int \sec t\, dt = \ln|\sec t + \tan t|$ (differentiate to check: $\frac{\sec t \tan t + \sec^2 t}{\sec t + \tan t} = \sec t$).

(5) $x_p = -\cos t \sin t + \big(\sin t - \ln|\sec t + \tan t|\big)\cos t = -\cos t\, \ln|\sec t + \tan t|$.

Check (write $\ell = \ln|\sec t + \tan t|$, $\dot{\ell} = \sec t$): $\dot{x}_p = \sin t\, \ell - 1$, $\ddot{x}_p = \cos t\, \ell + \sin t \sec t = \cos t\, \ell + \tan t$, so $\ddot{x}_p + x_p = \tan t$.

SECTION 10 • SUMMARY

10. Summary and lecture map

RESULT	STATEMENT	NOTES	ROBINSON	LECTURE
Existence and uniqueness	$f, \partial f/\partial x, \partial f/\partial \dot{x}$ continuous in a cube; data $x(t_0), \dot{x}(t_0)$	p. 5	Thm 11.1, p. 102	Sep 22
Superposition	$\alpha x_1 + \beta x_2$ solves $x'' + px' + qx = 0$	p. 5	Sec. 11.2, p. 103	Sep 22
General solution	$x = C_1 x_1 + C_2 x_2$ with x_1, x_2 independent	p. 5	Sec. 11.3, pp. 104–106	Sep 22
Wronskian	$W = x_1 x_2' - x_2 x_1'$; $W' = -pW$, $W = Ce^{-\int p}$: $W \equiv 0$ or never 0	p. 5	Sec. 11.4, Ex. 11.2	Sep 22
Constant coefficients	$k^2 + pk + q = 0$; Cases 01, 02, 03	p. 5	Ch. 12, Sec. 16.1	Sep 22–24
Mass–spring–damper	Over/critical/under-damped	p. 5 (spring)	Ch. 13	Sep 22
Inhomogeneous	$x = x_H + x_N$; educated guess; $x'' + x' - 6x = 12,\ 36t,\ 4e^{-2t}$	p. 6	Ch. 14	Sep 24–29
Resonance	Guess times t; damped peak at $\alpha^2 = \omega^2 - \lambda^2/2$	p. 6	Ch. 15	–
Reduction of order	$x = yu$ gives a first-order equation for $z = y'$	p. 7	Ch. 17	Sep 24–29
Variation of constants	$u_1' = -fx_2/W$, $u_2' = fx_1/W$	p. 8	Ch. 18	Oct 1

WHAT COMES NEXT

Topic 4 rewrites $\ddot{x} + p\dot{x} + qx = f$ as a first-order system for $(x, \dot{x})$ and replaces "two independent solutions" by a fundamental matrix, the Wronskian by a determinant of vector solutions, and the auxiliary equation by an eigenvalue problem.